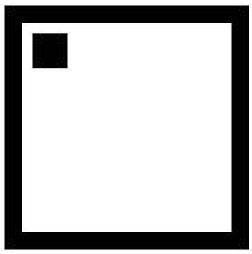
Alemeno Frontend Internship Assignment
Android Native App forCustom Marker Detection & Extraction

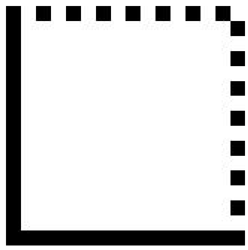
Problem Statement:

Design and implement a custom visual marker system - similar to a QR code - that is printable and machine-readable via a smartphone camera. The solution should be an Android native application made using React Native. The app should be capable of accessing the device camera, rendering a live camera feed, identifying the custom marker within the frame, isolating and extracting it, applying orientation correction, and displaying 20 processed markers from 20 different frames.

Details & Constraints:

1. You must use React Native to build the application
2. You can design your own custom marker or use any 1 of the 2 custom markers provided in the attached zip. Below are the 2 markers for reference. The measurements for these are also available in the zip file.





3. If you are designing your own custom marker, here are the constraints:
    a. You can only use black and white color
    b. The overall marker should be a square and NOT any other shape like circle or rectangle
    c. Minimum 60% of the inside area should be empty so information can be encoded inside the marker.
4. The detection logic should be robust enough that it doesn't falsely detect similar shapes or objects and only detects the correct marker. You can find some test images in the zip file. It should only detect the provided Correct Marker Images and should not detect the provided Incorrect Marker Images.
5. The live camera feed should be minimum 2000x2000px and maximum 3000x3000px
6. The processed 20 markers displayed at the end on UI should all be exactly 300x300px

Evaluation Criteria:

1. Speed - Ideal total scan-to-result time should be under 3000 ms; faster is better.
2. Orientation robustness - Orientation correction must work reliably across all rotations.
3. Extraction accuracy - The isolated marker should be tightly cropped with no surrounding padding and zero geometric skew.
4. Detection accuracy - should only detect the correct marker

Deliverables:

1. An installable APK file of the app.
2. Public repo with complete source code and setup instructions.
3. A PDF explaining the approach you took and why
4. If you have designed your own custom marker, share the measurements or generation logic for it and a few test images in different orientations.